What Are We Actually Benchmarking in Robot Manipulation?

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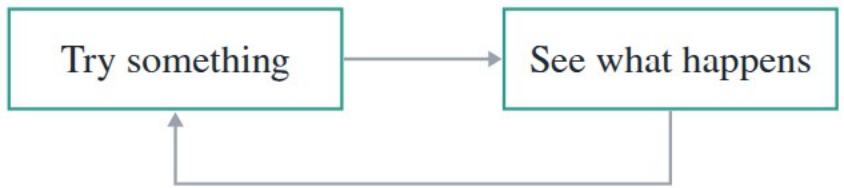




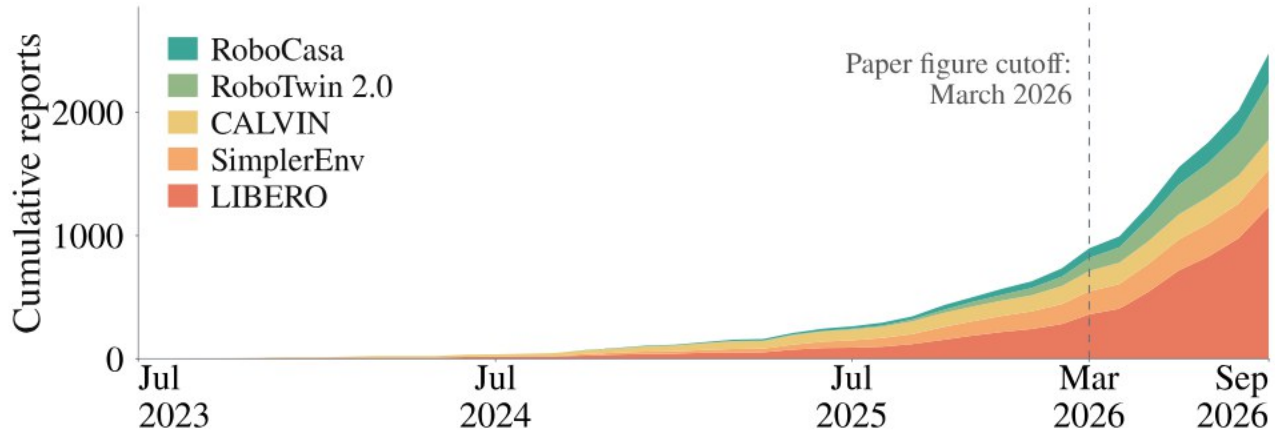


A manipulation benchmark has two roles

1. Testing and comparing methods



2. A leaderboard for judging progress



79 papers reported LIBERO results in March 2026 alone

The score decides what gets read/published

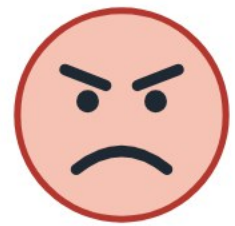
Paper 1



Reviewer 2

"Outperforms all baselines on LIBERO."

Paper 2



Reviewer 2

"No LIBERO results."

"Below state of the art on LIBERO."

A proxy for what we care about









Footage: Figure, Physical Intelligence, Generalist AI, DROID.

Four ways a higher score can mislead

1. A shortcut scores high
2. The gain is noise
3. The policy is tuned to the fixed test
4. Better data looks like generalization



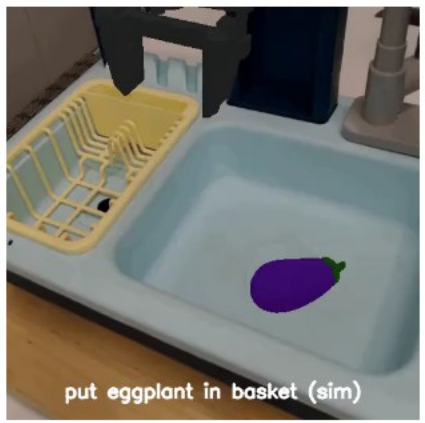
LIBERO

Liu et al., 2023



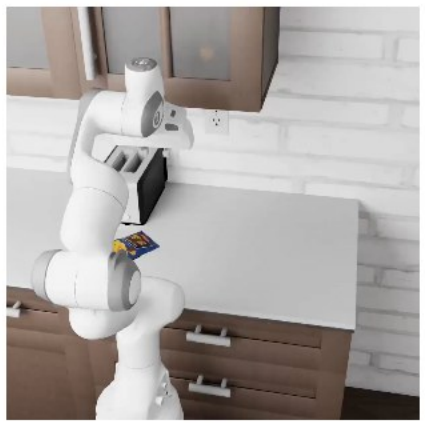
CALVIN

Mees et al., 2022



SimplerEnv

Li et al., 2025



RoboCasa

Nasiriany et al., 2024



RoboTwin 2.0

Chen et al., 2025

CHECK 1

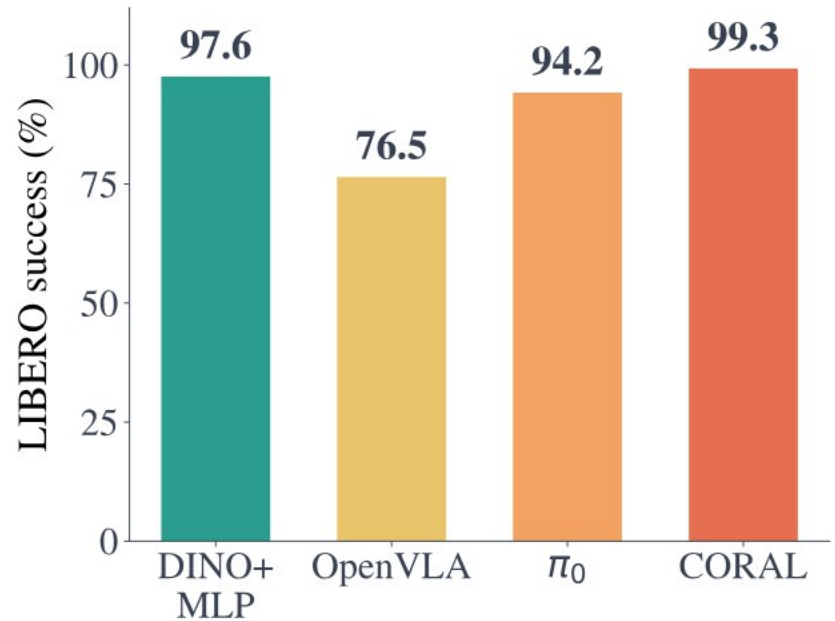
Is there a shortcut?

Shortcut: a way to score high without the ability the benchmark is meant to test

Common belief: a high LIBERO score needs robot pretraining and language understanding

Our policy has neither

- No robot pretraining
- Treats the instruction as an ID, not as words
- **0.09B** parameters (OpenVLA: 7B)



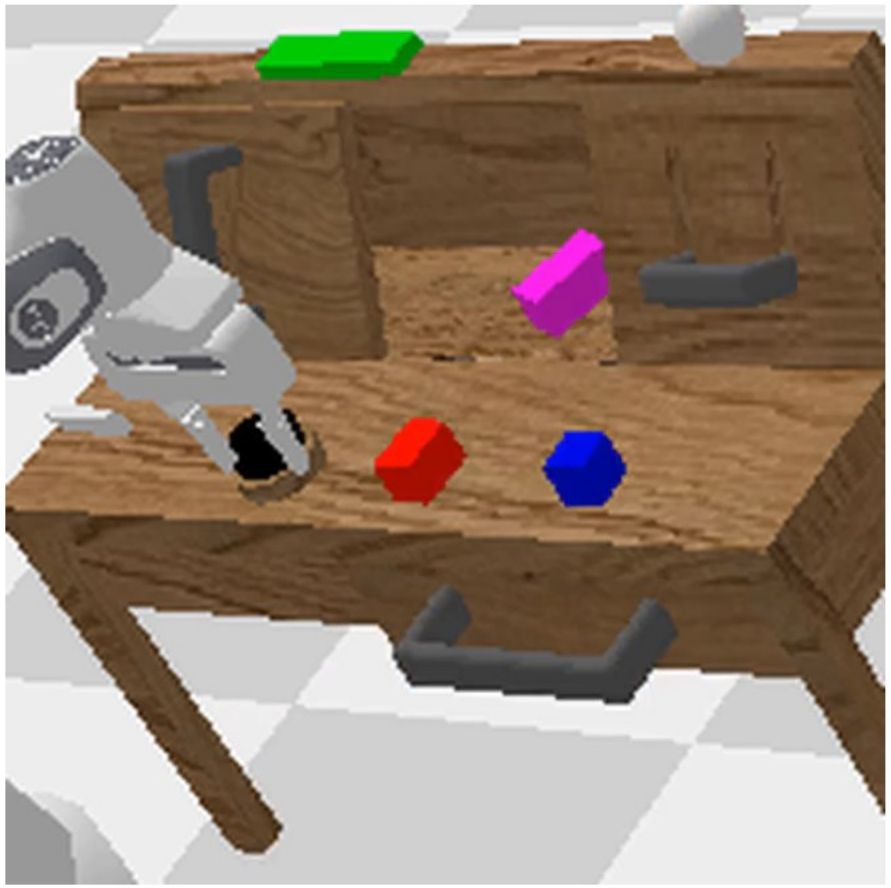
CALVIN: matches widely used policies, but not the best

CHECK 1

Why?



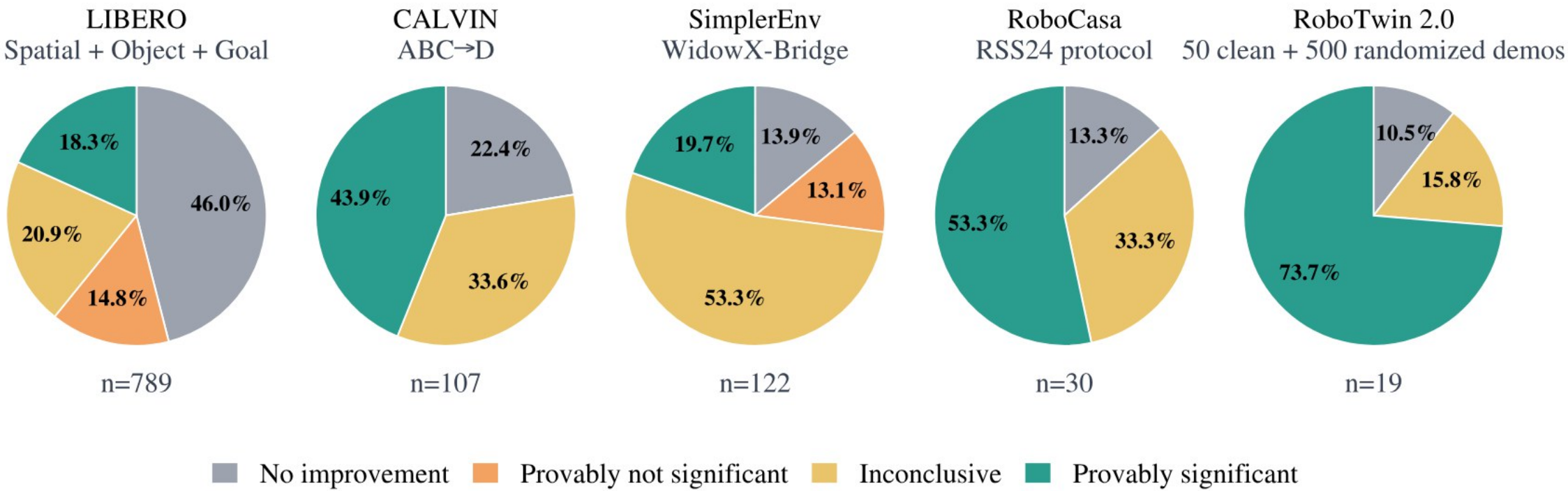
LIBERO-Object, one task: 50 training demos



CALVIN: 50 of the 1,000 test episodes

CHECK 2

Is a gain real, or noise?

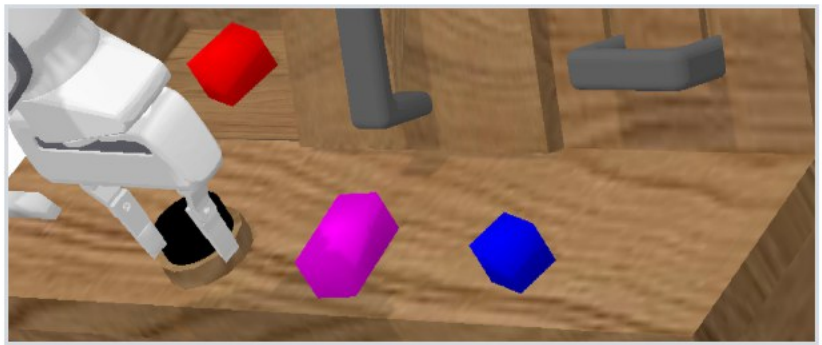


Through May 2026. Each paper vs. its reported previous best. Significance at 5%, from reported scores only.

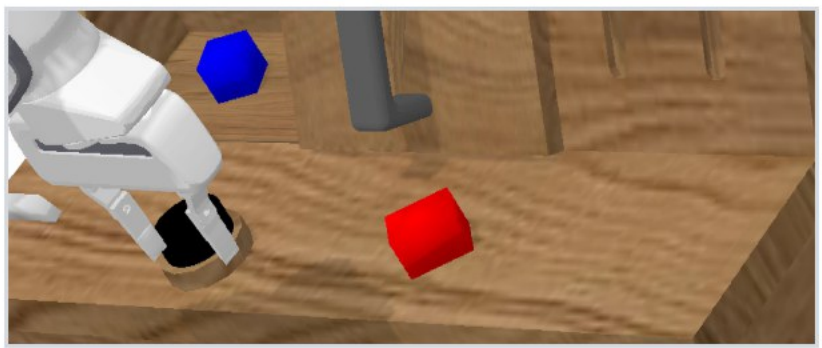
CHECK 3

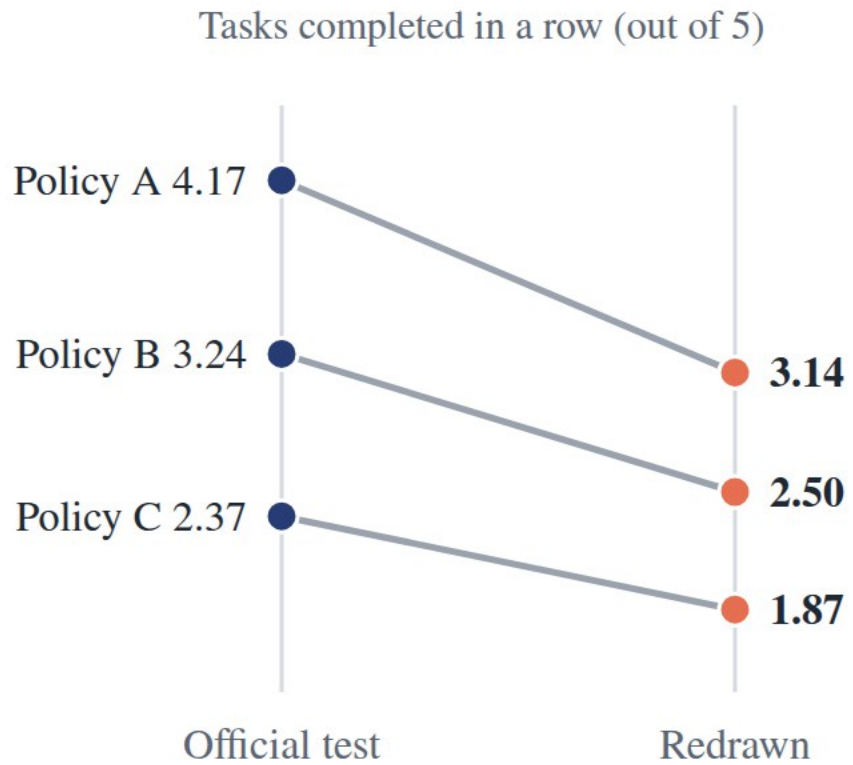
Does the score hold with small variation?

Official test: the same 4 spots



Redrawn from the training range





CALVIN ABC→D, scene D. 1,000 test episodes per policy and condition.

CHECK 4

Generalization, or better data?



Usual training data: real robot (Bridge)

⇔

real-to-sim gap



Test: simulation (SimplerEnv)

95.8%

X-VLA: 0.9B, broad real and sim pretraining

94.8%

Ours: 0.02B, no robot pretraining, scripted demos next to the test

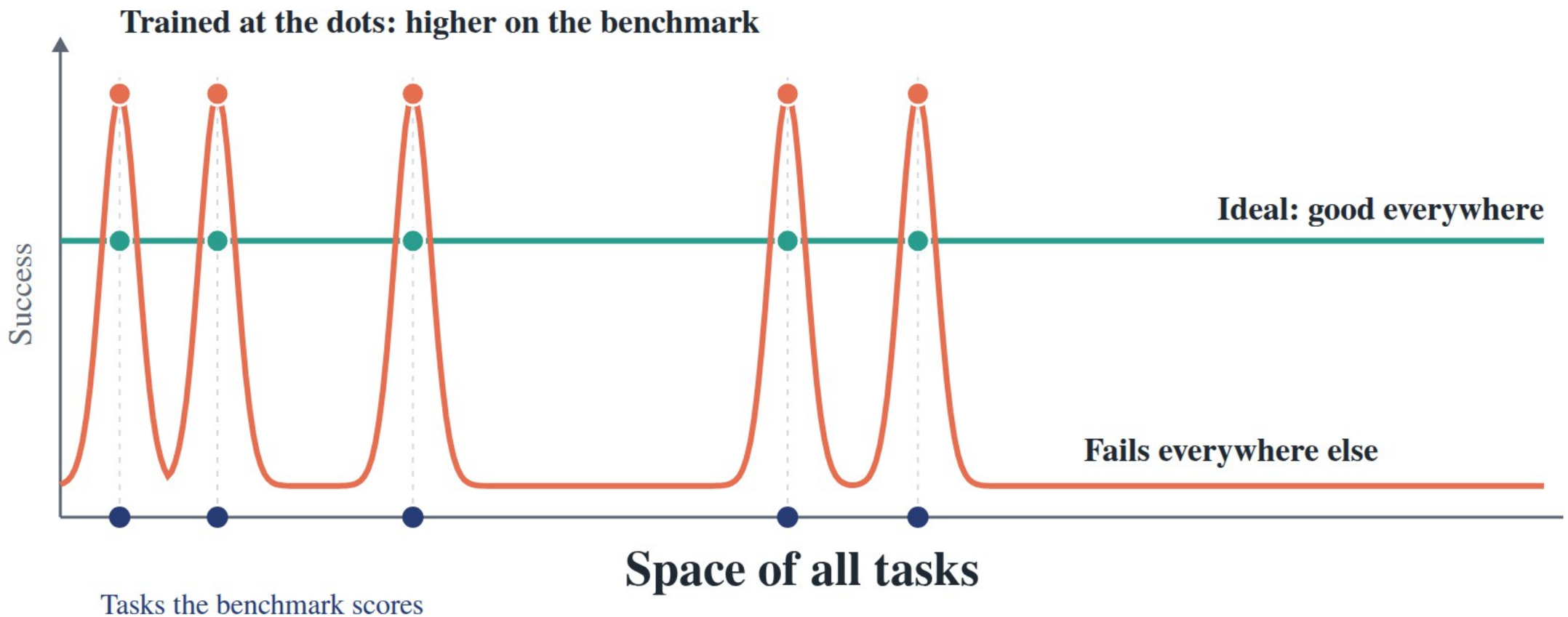
COMMON CAUSE

Most of these problems share one cause

A public, fixed, narrow test set

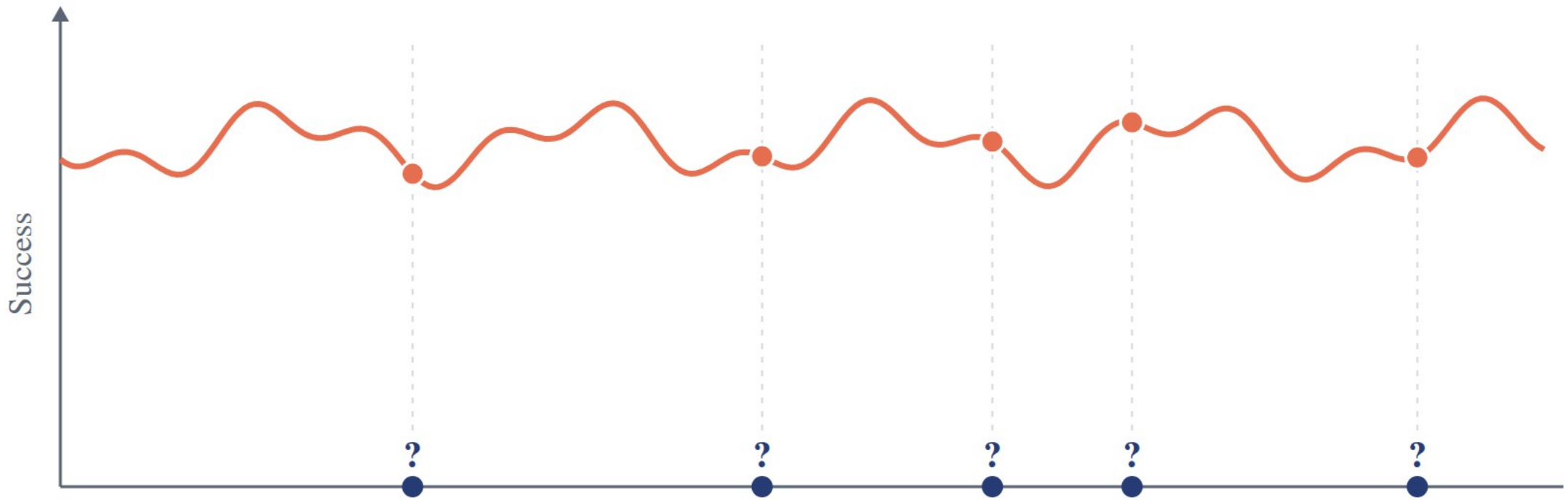
COMMON CAUSE

People can optimize to a fixed test set



FUTURE BENCHMARKS

1. Test after release

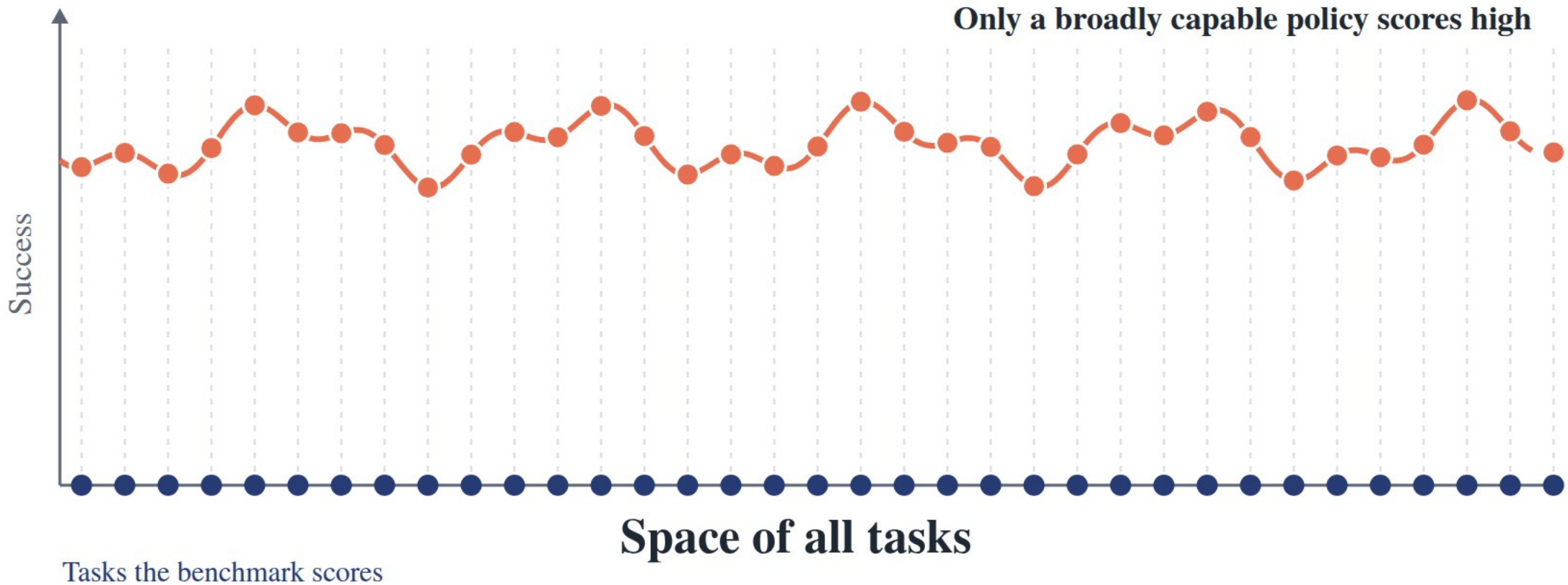


If the test is hidden, how do we know it is a good test?

RoboCurve does this on real robots. The BEHAVIOR Challenge ranks its finalists on hidden test scenes.

FUTURE BENCHMARKS

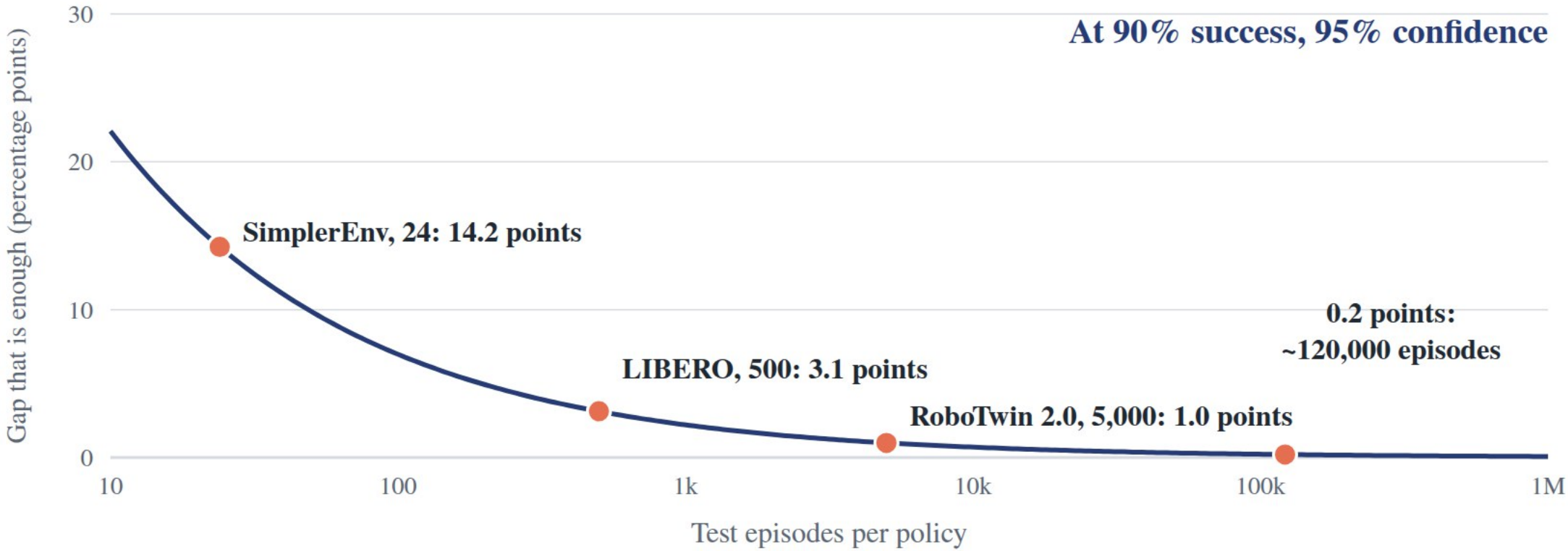
2. Or cover widely



Tasks the benchmark scores

FUTURE BENCHMARKS

3. Give clear guidance on what gap is enough



Episodes per reported score: SimplerEnv 24 per task, LIBERO 500 per suite, RoboTwin 2.0 5,000 per setting (50 tasks × 100). One-sided test at the 5% level, as in the paper.

Q&A

Paper

Website

Poster 116

Today, 5:00–6:30 PM

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